

Optimal Placement of Electric Vehicle Charging Station with V2G Provision using Symbiotic Organisms Search Algorithm

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Abstract—Electric Vehicles (EVs) are one of the most efficient technologies for the sustainable transport sector in the present world considering its almost nil carbon emission. To cope up with the ever-increasing demand of power for EV charging, the distribution system's power loss increases and significant changes in voltage profile occurs. Improper installation of the EV Charging Stations (EVCS) significantly reduces the power quality of the distribution network, impacting its efficient process. In this paper, the installation of EVCS in the IEEE 33 bus radial distribution system is being outlined. The distribution system has been split into three areas in order to reach charging facilities in various locations within an area and it has been ensured that one charging station is allocated in each area. The issue is being categorized as the optimization problem and it has been dealt using Symbiotic Organisms Search Algorithm (SOS) for finding the optimal location of EVCS.

Keywords—Distribution System, Electric Vehicle (EVs), Electric Vehicle Charging Stations (EVCS), Real Power Loss Minimization, SOS.

I. INTRODUCTION

The most challenging concerns in the present generation are air pollution, global warming threat, and environmental issues [1]. Environmental awareness towards alarming emission of greenhouse gases, financial gains in terms of fuel use, and the incentives offered by the government are the key attributes for changing the mindset of people from conventional vehicles to EV [2]. The use of EVs is a technique to replace conventional fossil fuel-powered vehicles which helps to reduce carbon emissions and fossil fuel utilization [3]. For EV owners to charge their vehicles, EVCS is a basic necessity. The provision of charging infrastructure is inevitable for widespread EV implementation [4]. Thus, the proper allocation of EVCS is necessary for the distribution network by keeping the real power loss minimum with a proper voltage profile. The inappropriate positioning of EVCS will lead to additional energy consumption which creates a negative consequence on the power system network's energy requirements.

The authors in [5] have used GWO and WOA optimization tools to solve EVCS allocation issues in IEEE 33 bus distribution network. In [6], the authors have implemented parking lot allotment using GA in the distribution network. The authors in [7] have assigned the charging station in a distribution network that overlaps with a traffic network in order to reduce power loss and voltage fluctuation. In [8]-[10], most of the research works focus on the power loss

minimization for optimally allocating EVCS using different soft computing techniques.

The proposed research emphasizes on optimally allocating EVCS in IEEE 33 bus radial distribution system. The distribution network is separated into three areas for widespread access to EVCS. V2G is an exceptional attribute that improves power system quality. Both G2V and V2G provisions have been considered. In this research paper, the SOS algorithm has been implemented for allocating EVCS at suitable positions where the distribution network's active power loss is minimized along with maintaining a healthy voltage profile.

The research paper is arranged as follows: Section II deals with problem optimization where the real power loss minimization is the objective function along with constraints. Section III includes the SOS optimization tool applied to the system which includes a comprehensive assessment of the system considering various scenarios for optimally allocating the best position for EVCS. Section IV deals with results obtained from the cases considered to examine the impacts on power system parameters. Section V depicts the concluding remarks of this research paper.

II. PROBLEM OPTIMIZATION

A. Objective Function Formulation

For this research paper, optimization of single objective problem has been computed for minimizing the real power loss.

Here, we have considered the w^{th} branch of the distribution network.

z^w = Impedance of the w^{th} branch.

r^w = Resistance of the w^{th} branch.

x^w = Reactance of the w^{th} branch.

i^w = Current flowing in the w^{th} branch.

The impedance z^w can be divided into its real and imaginary part as,

$$z^w = r^w + jx^w \quad (1)$$

Hence, the total apparent power loss s^w is given by,

$$s^w = (i^w)^2 \times z^w \quad (2)$$

Evaluation of the entire load in the network is computed by,

$$t_{load} = \sum_{ws=1}^{Nws} p_{ex}^{ws} + p_{evcs}^{ws} \quad (3)$$

Where, t_{load} = Entire load of the network.

Nws = The overall number of buses within the network.

p_{ex}^{ws} = The existing load in ws^{th} bus.

p_{evcs}^{ws} = The extra load because of EVs at charging stations which is linked to ws^{th} bus.

p_{evcs}^{ws} can be further simplified as,

$$p_{evcs}^{ws} = (n_{ev}^{g2v} \times r_{chg}) - (n_{ev}^{v2g} \times r_{dis}),$$

if $ws \in$ the location of charging station.

$$= 0, \text{ if } ws \notin \text{ the location of charging station.} \quad (4)$$

Where, n_{ev}^{g2v} = The overall number of EVs linked to the charging stations in G2V configuration.

n_{ev}^{v2g} = The overall number of EVs linked to the charging stations in V2G configuration.

r_{chg} = EVs rate of charge.

r_{dis} = EVs rate of discharge.

EVCS will be installed on that particular bus where the extra load for the charging stations will be connected. The bus number ($ws \notin$ location of charging station) serves as the deciding parameter in this paper for optimization.

Hence, the total apparent power loss is divided into its real and reactive power loss component.

$$pl^w = \text{re}(s^w) \quad (5)$$

$$ql^w = \text{img}(s^w) \quad (6)$$

Where, pl^w = Real power loss component in w^{th} branch.

ql^w = The loss component of reactive power in w^{th} branch.

In radial distribution network, the overall loss of real power, pl is being evaluated by,

$$pl = \sum_{w=1}^{nw} pl^w \quad (7)$$

Where, nw = The overall number of branches.

Thus, objective function is being represented as,

$$\min(f) = \min(pl) \quad (8)$$

B. Constraints

1) Voltage Limit Constraints: At any time interval, the voltage of each node should be within the acceptable limit as

represented in (9).

$$v_{\min} \leq v_{ws} \leq v_{\max}, \text{ for } ws = 1 \text{ to } Nws \quad (9)$$

Where, $v_{\min}$ = The minimum limit of voltage.

$v_{\max}$ = The maximum limit of voltage.

v_{ws} = Bus voltage in ws^{th} bus.

2) Current Flowing Limit Constraints: Current passing through each branch (i_w) must be lower than the maximum permissible level ($i_{\max}$).

$$i_w \leq i_{\max}, \text{ for } w = 1 \text{ to } nw \quad (10)$$

Where, i_w = Current flowing in the w^{th} branch.

$i_{\max}$ = Maximum current flowing limit.

nw = Total number of branches.

3) Load Balancing Constraints: The power delivered from the substation should be equivalent to the total of the demand and the loss.

$$p_{ss} = \sum_{ws=1}^{Nws} p_{ws} + pl \quad (11)$$

Where, p_{ss} = The power provided by the substation.

p_{ws} = The load on ws^{th} bus.

Nws = The overall number of buses in the system.

pl = The total real power loss.

III. OPTIMIZATION TOOL

Min-Yuan Cheng and Dobby Prayogo proposed the SOS algorithm in 2014 [11]. SOS is a powerful and robust metaheuristic algorithm which has been implemented to solve optimization and complex engineering design issues. This algorithm mimics the symbiotic interaction techniques that organisms adapt for survival in the nature. The SOS algorithm resembles the real-world biological relationship in three phases i.e., mutualism, commensalism, and parasitism. Mutualism refers to the symbiotic interaction among two different species in which both the organisms is benefitted. Commensalism refers to the symbiotic interaction among two different organisms where one benefits while the other one is unharmed. Parasitism refers to the symbiotic interaction among two different organisms where one is benefitted while the other one is actively affected [12]. The SOS algorithm has been implemented in this research paper for searching the optimal nodes for the allocation of EVCS.

For optimal allocation of EVCS, the proposed flow chart is being presented in fig. 1. In this research paper, the search agent denotes the location of random nodes where EVCS will be connected. For calculating the objective function, we have used the distribution system's backward forward sweep load flow technique along with various constraints. The locations of EVCS will be updated according to the SOS optimization method.

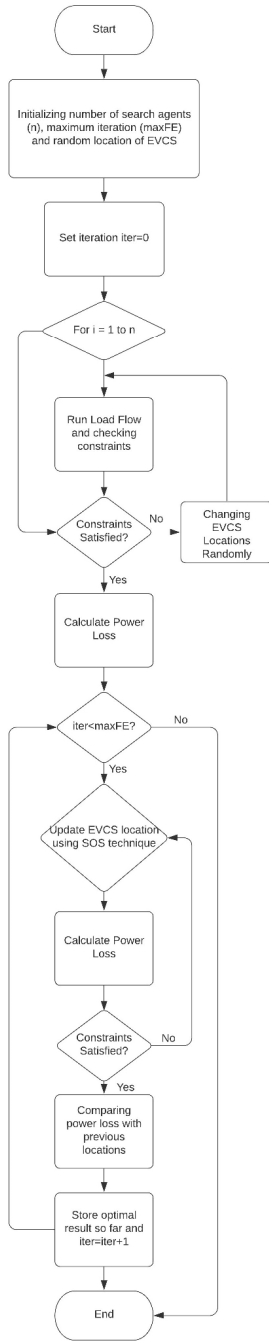


Fig. 1. Optimal position allocation flow chart by applying SOS Algorithm

IV. RESULTS AND ANALYSIS

The real power loss as well as other parameters of power system of the distribution network has been calculated using backward forward sweep load flow. We have considered charging rate as 19 kW [2] in G2V configuration and discharging rate as 8 kW [13] in V2G configuration. The authors have assumed that 20% EV can be operated in V2G configuration. We have illustrated two scenarios in this paper. Firstly, we have allocated the EVCS without separating the distribution network into different areas. Secondly, we have allocated the EVCS into three different areas after dividing the distribution network. For an instant of time, the problem has been solved. At the charging point, four instances have been taken into account: 10, 20, 30, and 40 EVs are linked at that point of time. A bad case scenario has been considered in

which the number of arrived EVs in the system cannot exceed 40 at a particular time. The EVCS has been allocated based on these scenarios.

According to IEEE 33 bus radial distribution network, the nominal loads have been connected at different nodes. The extra load due to EV charging has been taken into account at that node where the EVCS is assigned. The number of EVs along with charging rate and discharging rate determines the load to be connected in V2G and G2V configurations.

To perform this research, the MATLAB R2018b programming language is being chosen, installed on a system having an Intel i5 processor, having clock speed of 2.40 GHz, and RAM of 8 GB.

A. Allocation of EVCS without Area Division (Case-1) :

TABLE I. REAL POWER LOSS AND OPTIMAL POSITIONS IN CASE-1

Number of EVs	Optimal Positions			Real Power Loss in kW
10	2	19	20	211.8787
20	2	19	20	212.9523
30	2	19	20	214.0813
40	2	19	20	215.3023

The optimal EVCS locations for various numbers of EV penetration have been illustrated by using SOS in table I. The positions remain the same despite the variation in EVs count. This validates the applied algorithm's robustness. For various numbers of EVs, the voltage profile has been illustrated in fig. 2-5. Since the EVCS are placed relatively close to the substation, the variation in voltage profile is very less even when the number of EVs increases. The convergent curves for case-1 using SOS have been shown in fig. 6-9.

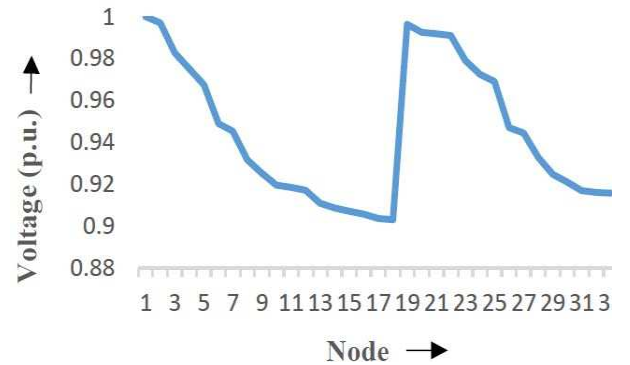


Fig. 2. The voltage profile with 10 EV for case-1

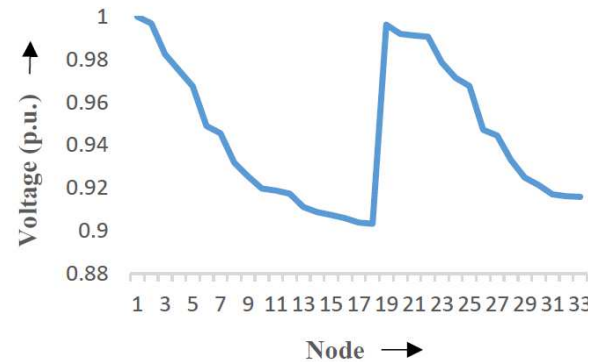


Fig. 3. The voltage profile with 20 EV for case-1

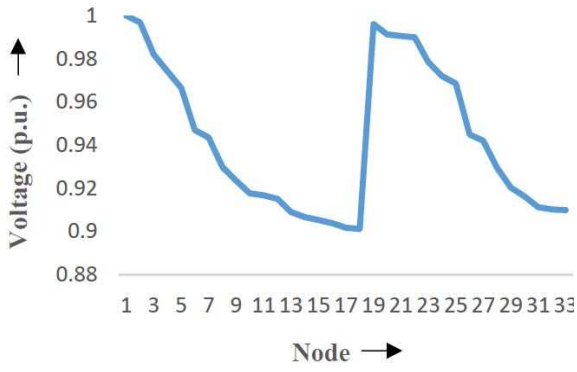


Fig. 4. The voltage profile with 30 EV for case-1

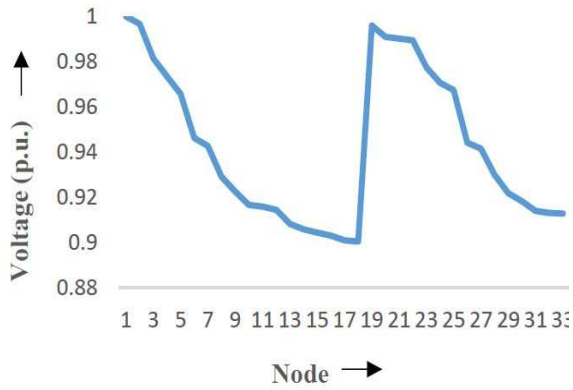


Fig. 5. The voltage profile with 40 EV for case-1

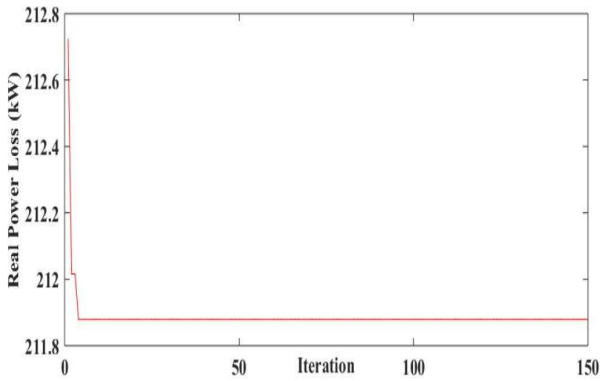


Fig. 6. Convergent Curve for case-1 with 10 EVs

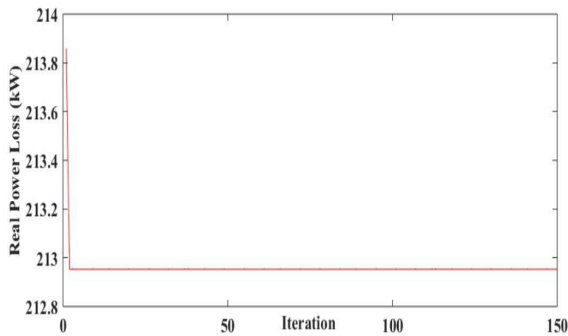


Fig. 7. Convergent curve for case-1 with 20 EVs

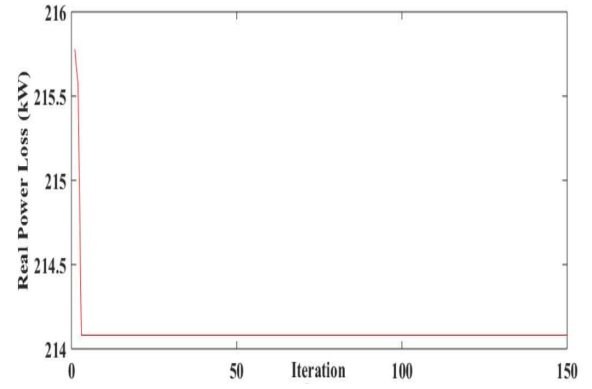


Fig. 8. Convergent curve for case-1 with 30 EVs

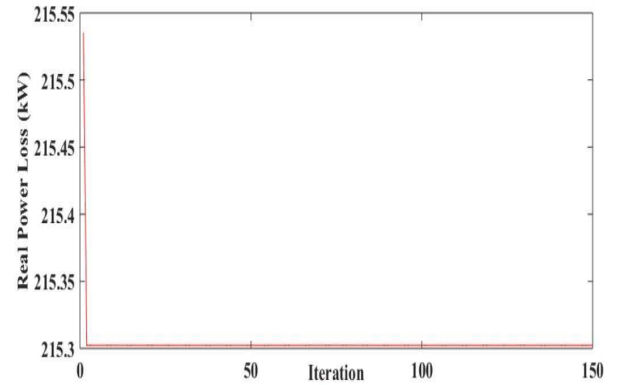


Fig. 9. Convergent curve for case-1 with 40 EVs

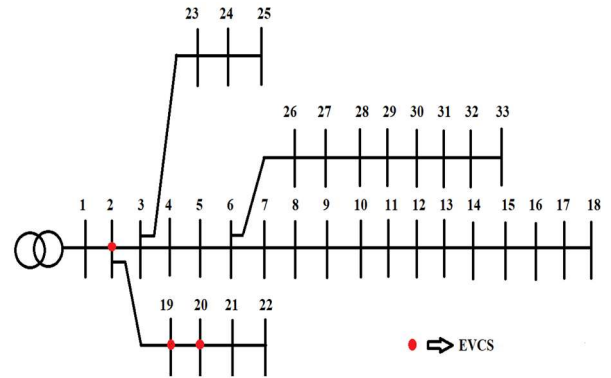


Fig. 10. Optimal position of EVCS for case-1

Best position of EVCS can be observed in fig. 10 without area division. The positions of EVCS are close to one another and it is possible that a large area remains unserved and EV owner has to travel a longer distance to reach a EVCS which is not feasible. Due to these issues, case-2 has been implemented which makes the scenario more realistic as EVCS will be divided on the basis of area and it will provide wider access to different areas.

B. Allocation of EVCS with Division of Area (Case-2) :

In distribution network, the area is being separated into three parts as illustrated in fig. 11 in order to resolve the issues persisted in case-1. The EVCS should be installed based on the needs of area. Otherwise, a consumer will have to go a considerable range to charge the EV. The division of area has

been done on the basis of assumed demographical conditions and charging demand as represented in fig. 11. Bus 1 has been assumed as the reference bus. For overcoming EV density, it is considered that each division has included crowded section. For widespread charging provision to the consumers, the EVCS has been assigned in each area.

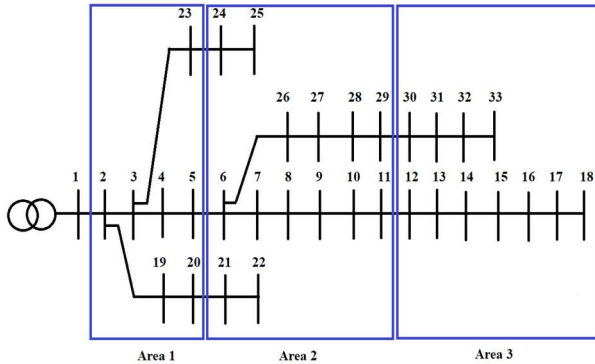


Fig. 11. IEEE 33 bus distribution networks area division

TABLE II. REAL POWER LOSS AND OPTIMAL POSITIONS IN CASE-2

Number of EVs	Optimal Positions			Real Power Loss in kW
10	2	21	30	214.5942
20	2	21	30	222.1786
30	2	21	30	228.542
40	2	21	30	235.5457

In Table II, the optimal location of EVCS for various number of EV have been illustrated. In fig. 12-15, various number of EVs voltage profile has been represented. The convergent curve for case-2 have been illustrated in fig. 16-19.

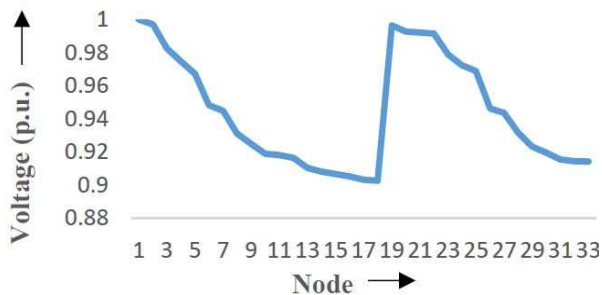


Fig. 12. The voltage profile with 10 EV for case-2

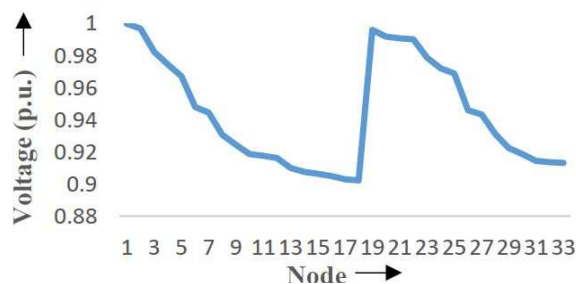


Fig. 13. The voltage profile with 20 EV for case-2

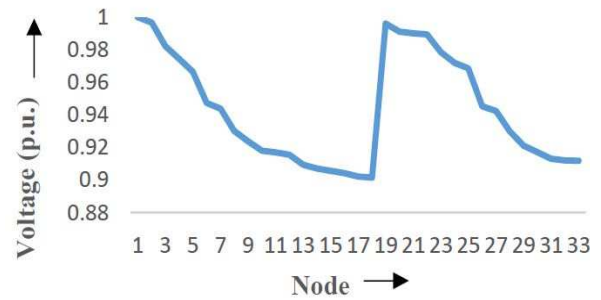


Fig. 14. The voltage profile with 30 EV for case-2

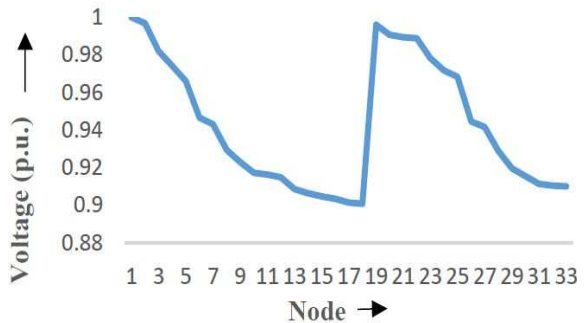


Fig. 15. The voltage profile with 40 EV for case-2

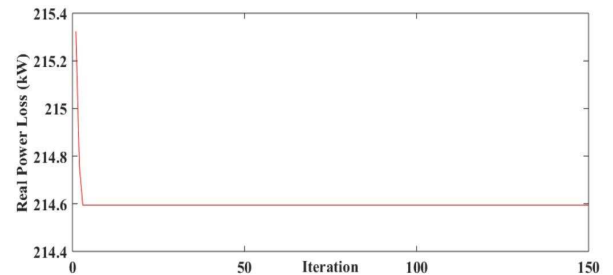


Fig. 16. Convergent curve for case-2 with 10 EVs

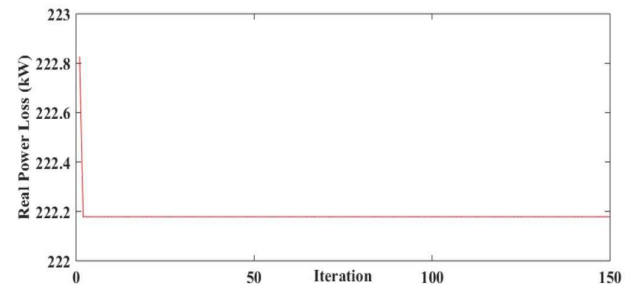


Fig. 17. Convergent curve for case-2 with 20 EVs

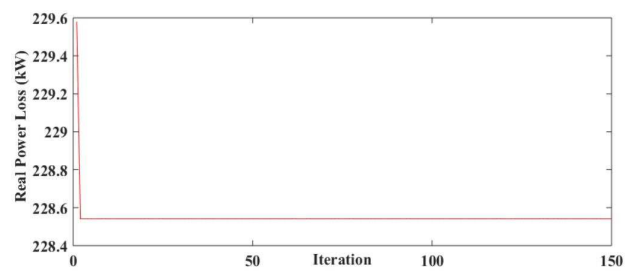


Fig. 18. Convergent curve for case-2 with 30 EVs

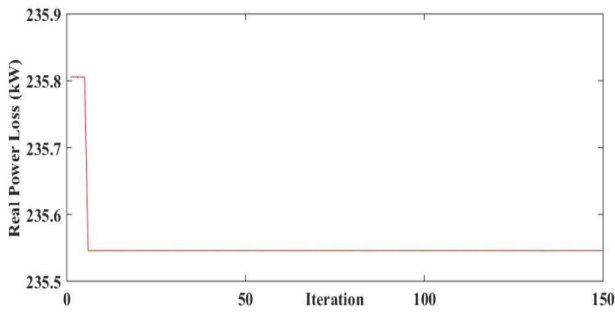


Fig. 19. Convergent curve for case-2 with 40 EVs

The positions of EVCS are more widespread which have been represented in fig. 20.

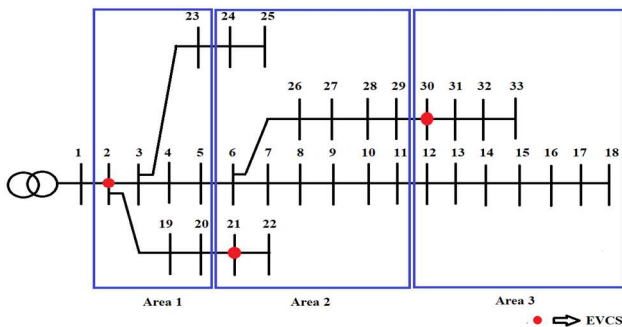


Fig. 20. Optimal position of EVCS for case-2

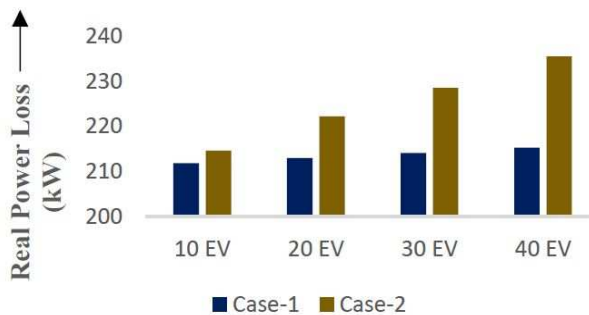


Fig. 21. Real power loss assessment between case-1 and case-2

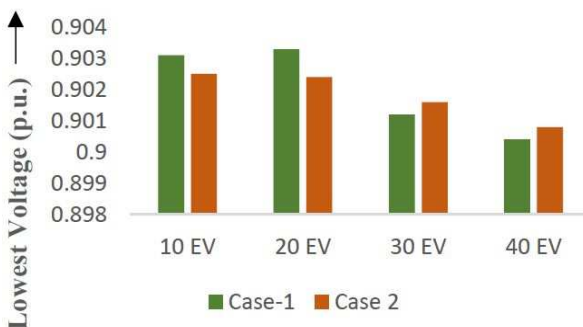


Fig. 22. Lowest voltage assessment between case-1 and case-2

In case-2, the real power loss is higher as compared to case-1 which have been illustrated in fig. 21 for similar number of EV penetration. The assesment of distribution networks lowest voltage have been represented in fig. 22.

V. CONCLUSIONS

The inadequate number of EVCS is the most significant barrier in acceptance of EV on large scale. This paper proposes a realistic solution on optimal placement of EVCS considering the charging stations are widely distributed for maximum accessibility as it is a consumer-oriented service. The SOS algorithm has been applied to solve the allocation issues and the obtained results shows that the solution is optimal and robust. It has been noticed that the placement of EVCS has a significant impact on the distribution network. Real power loss and level of voltage drop minimization should not be the only objective but providing EVCS access to widespread area is much more feasible and realistic.

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